LocalPoint3D · Far-OOD test

Run far_ood_2026-11-07_1947 · PointLLM 7B v1.2 · grey · seeds [0, 1, 2] · 42 scans

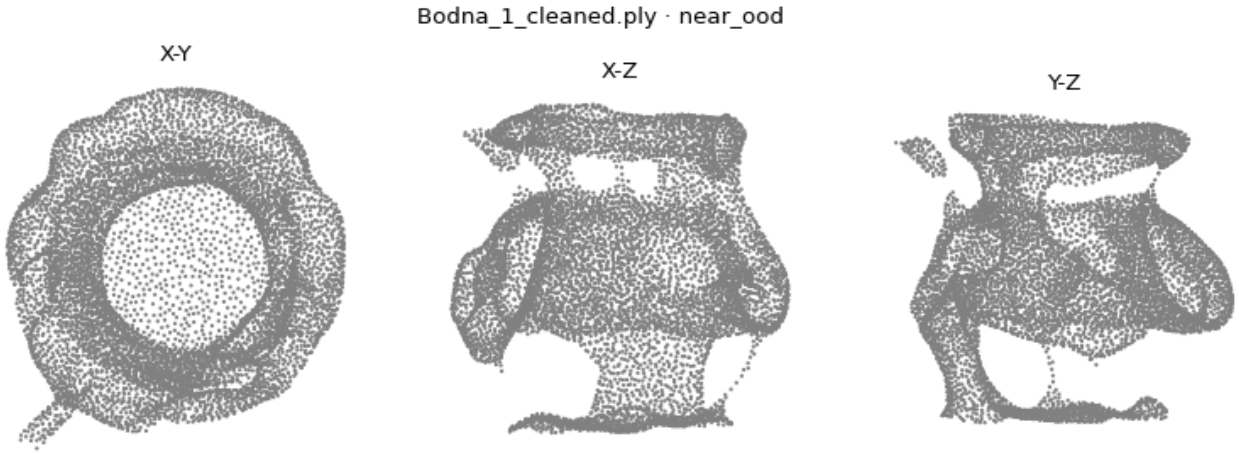
1. Summary

- Scans: **everyday** 20, **near_ood** 14, **far_ood** 8.
- Median fingerprint distance to the everyday scans: everyday 0.45, near_ood 0.77, far_ood 0.63. **Not** in the expected order.
- Multiple choice (own 4 options, 25 % by guessing), real scans correct: everyday 18/20, near_ood 0/14, far_ood 0/8.
- All-labels test (~1 in 39 by guessing), calibrated top-1 / top-3: everyday 5/20 / 12/20, near_ood 1/14 / 5/14, far_ood 0/8 / 0/8.
- How sure PointLLM is about its best label (median share): everyday 0.84, near_ood 0.53, far_ood 0.37.
- Far-OOD objects were called: rickshaw -> bench; sheel_pata -> faucet; thela_gari -> bench; thela_gari -> low stool; thela_gari -> bench; thela_gari -> low stool; thela_gari -> bench; tubewell -> bench.

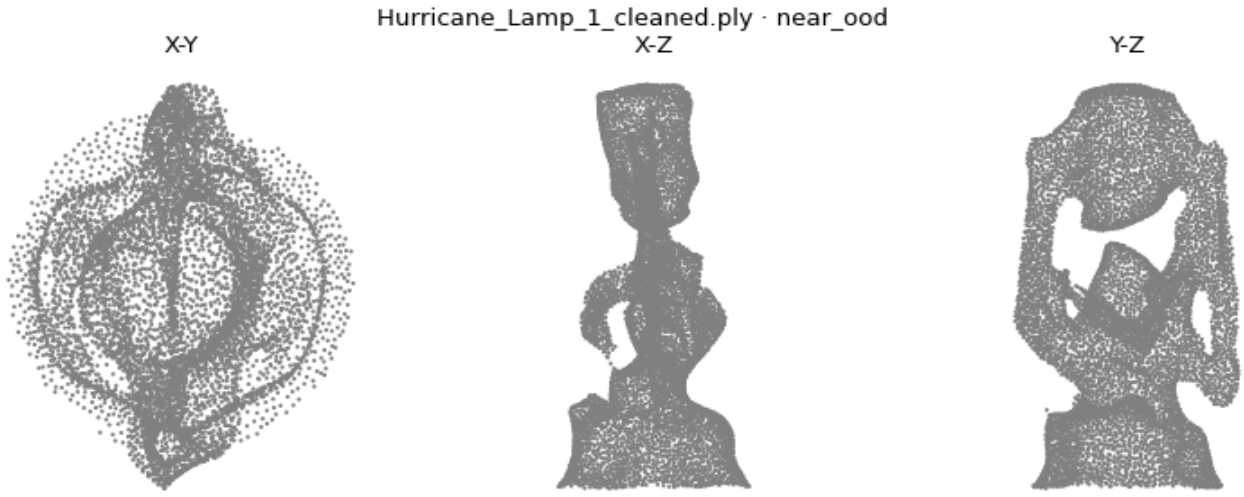
2. What we ran

group	scans	objects
everyday	20	chair, cup, keyboard, table, wooden_benched
near_ood	14	bodna, hurricane_lamp, kula, matir_cup, muura, patil, piri, rehal
far_ood	8	rickshaw, sheel_pata, thela_gari, tubewell

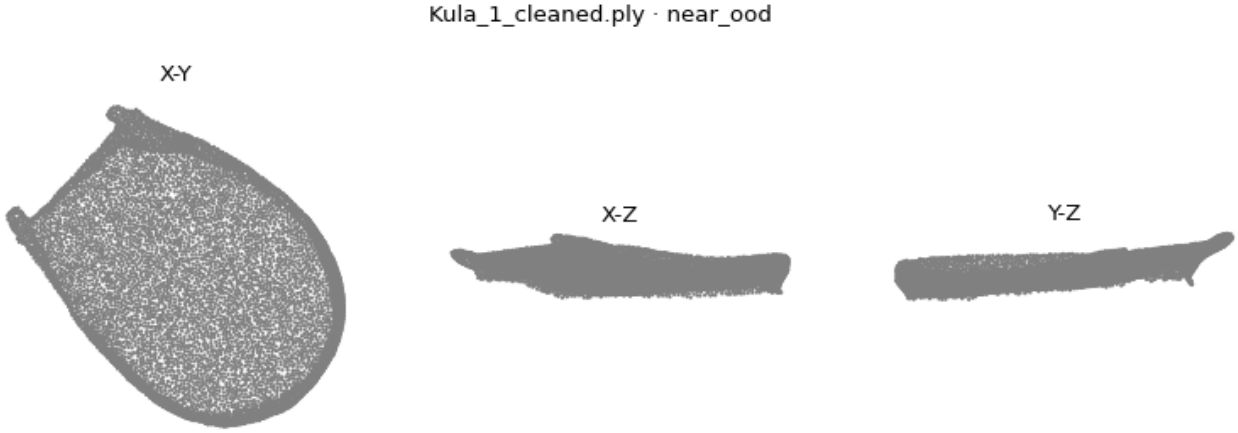
Model RunsenXu/PointLLM_7B_v1.2 (T4, float16). Input: 8192 points, grey, specks removed. Question: "What is this?". Answers are scored by likelihood, not generated. Multiple-choice seeds [0, 1, 2]; all-labels seeds [0]; blind inputs ['sphere', 'noise']. Options: data/mcq_options.csv.



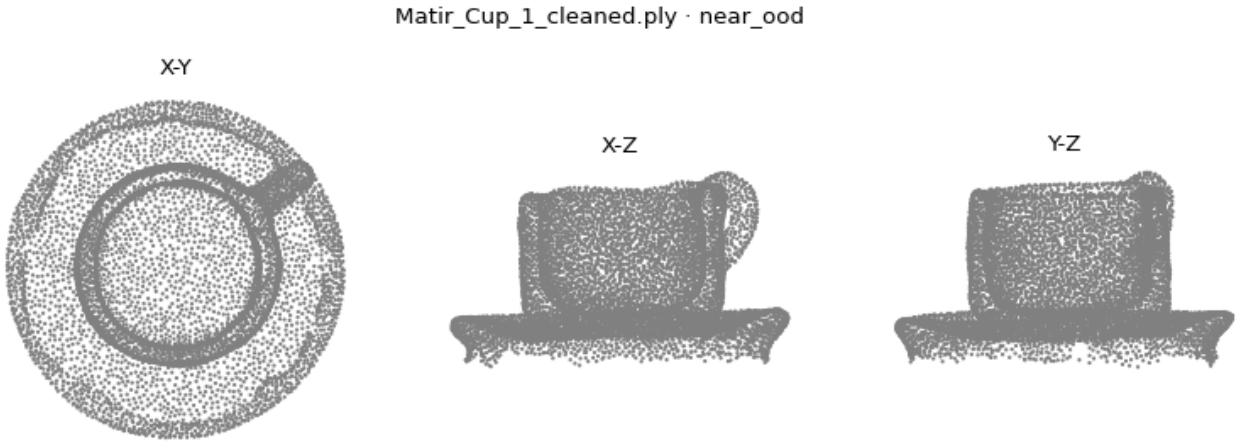
bodna (near_ood): what the model sees



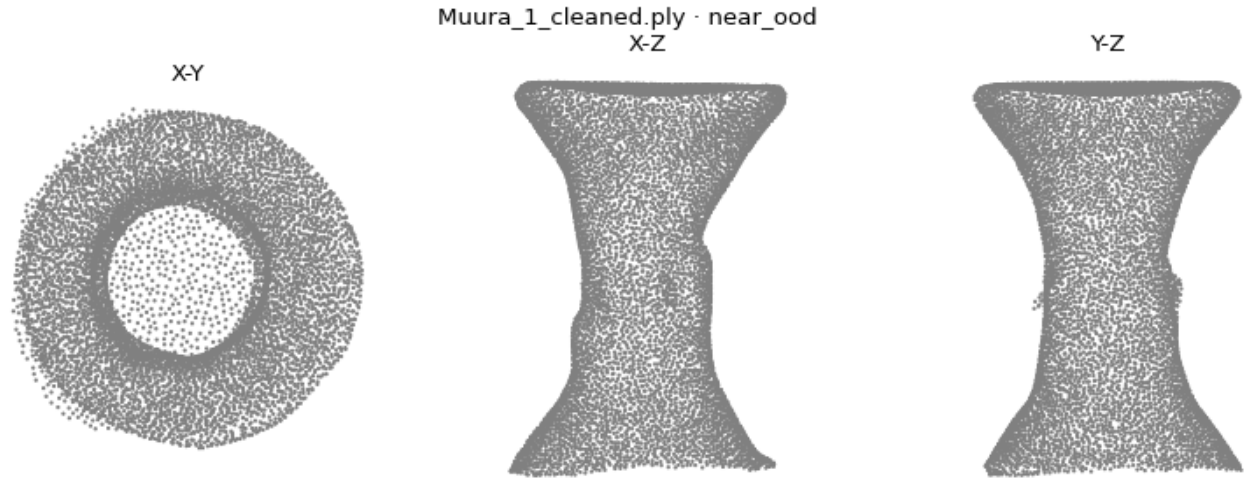
hurricane_lamp (near_ood): what the model sees



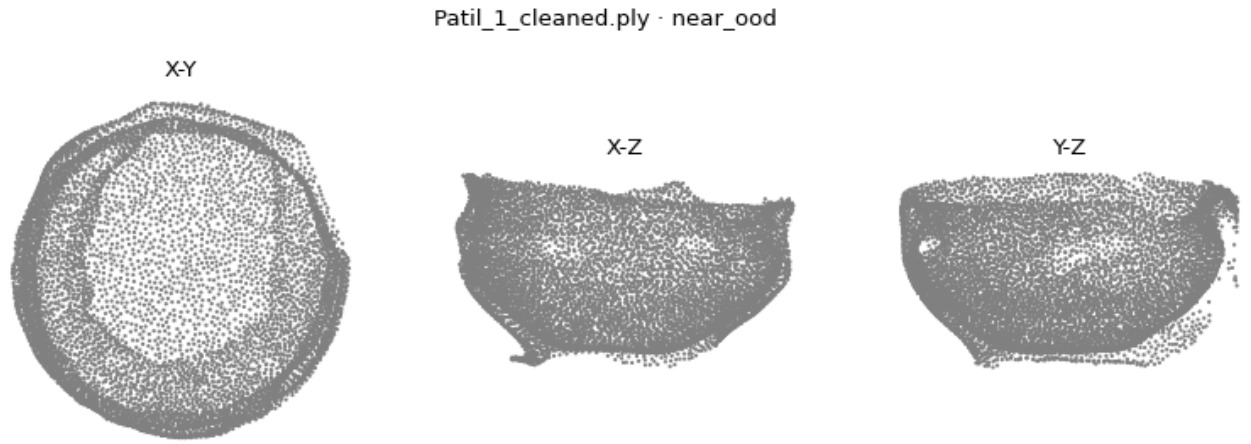
kula (near_ood): what the model sees



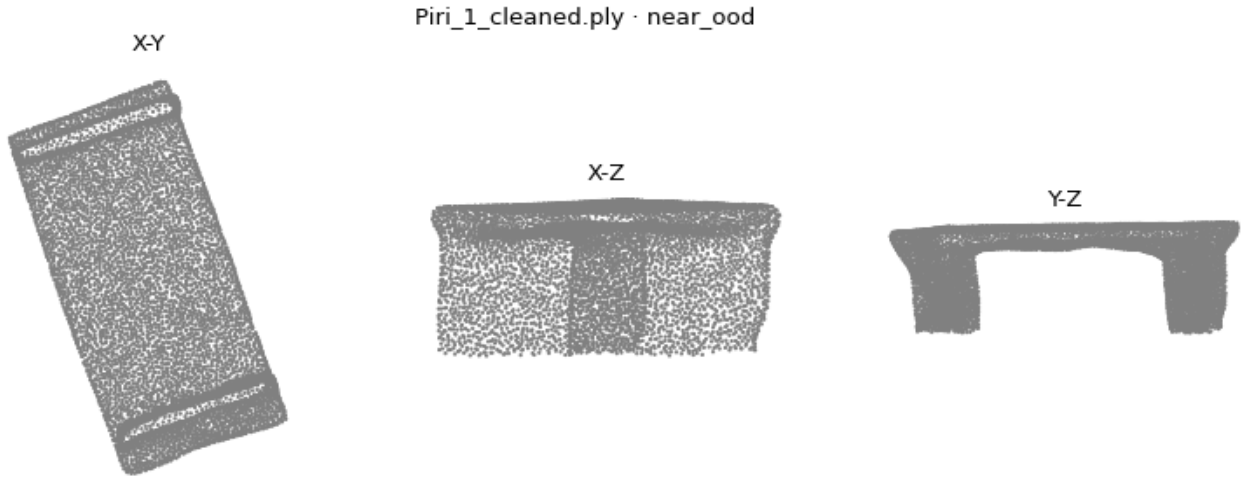
matir_cup (near_ood): what the model sees



muura (near_ood): what the model sees



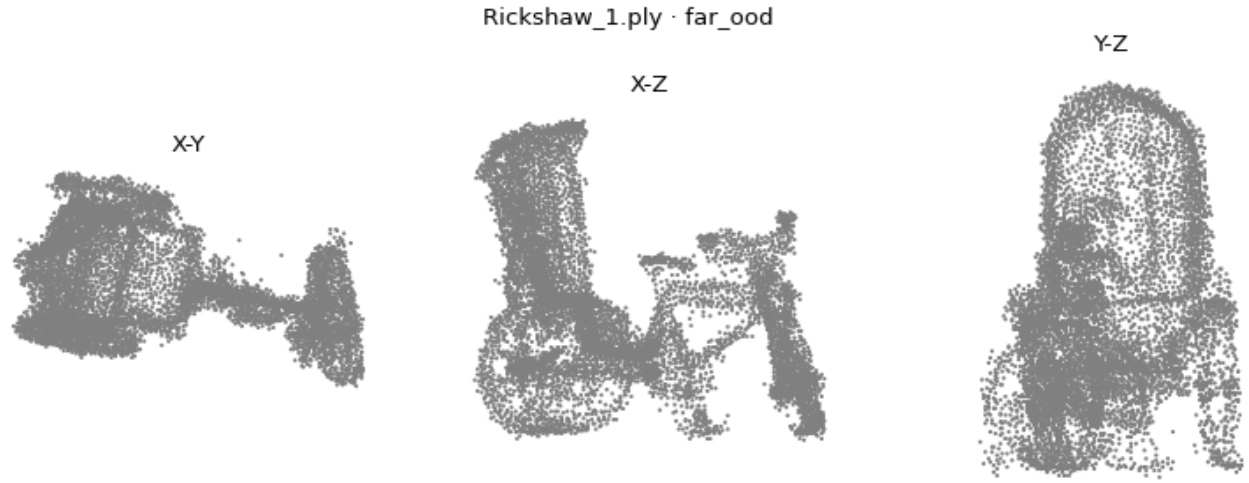
patil (near_ood): what the model sees



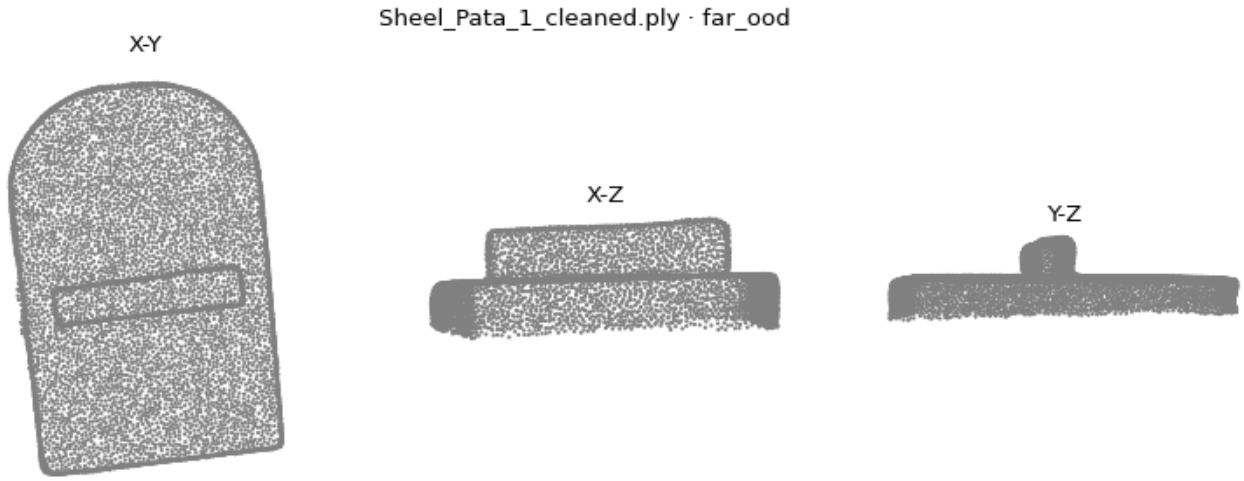
piri (near_ood): what the model sees



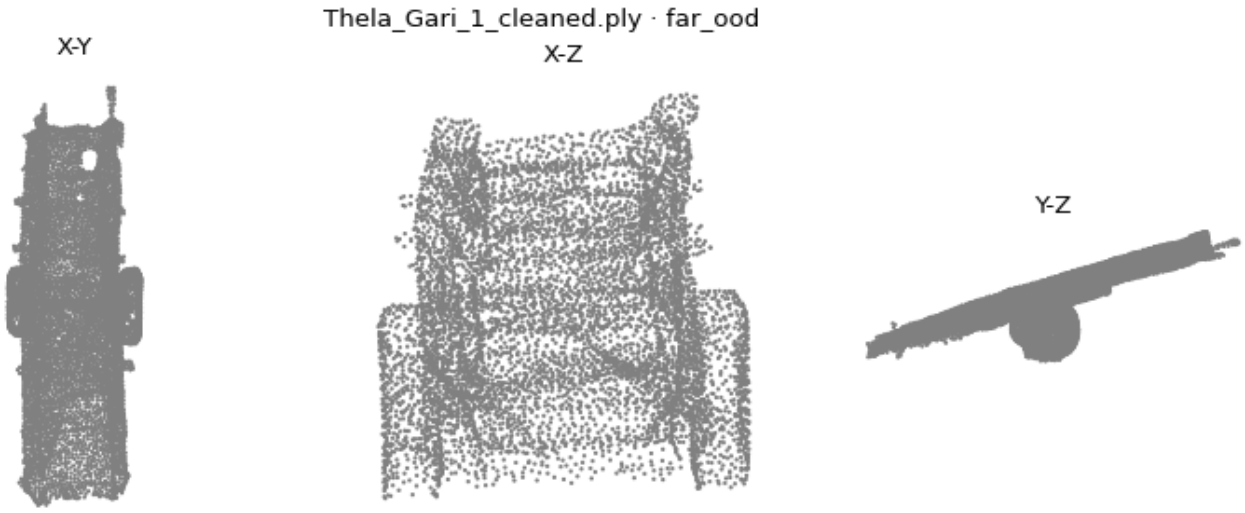
rehal (near_ood): what the model sees



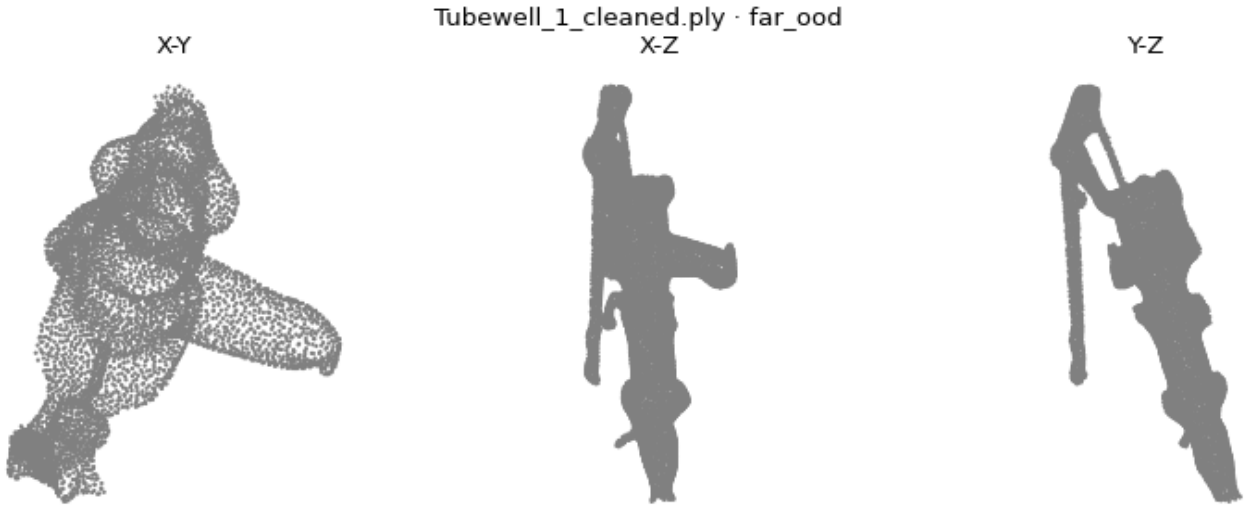
rickshaw (far_ood): what the model sees



sheel_pata (far_ood): what the model sees

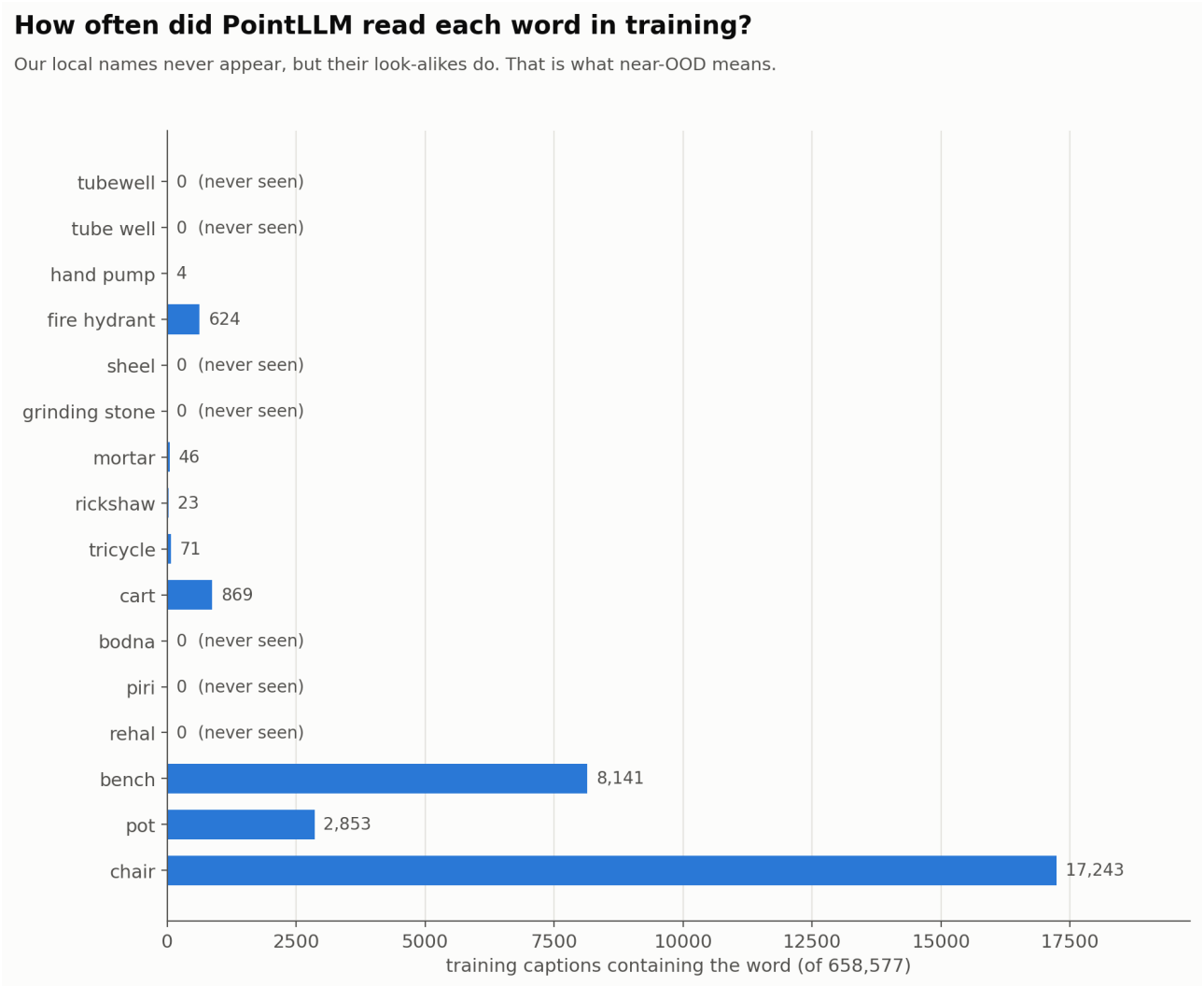


thela_gari (far_ood): what the model sees

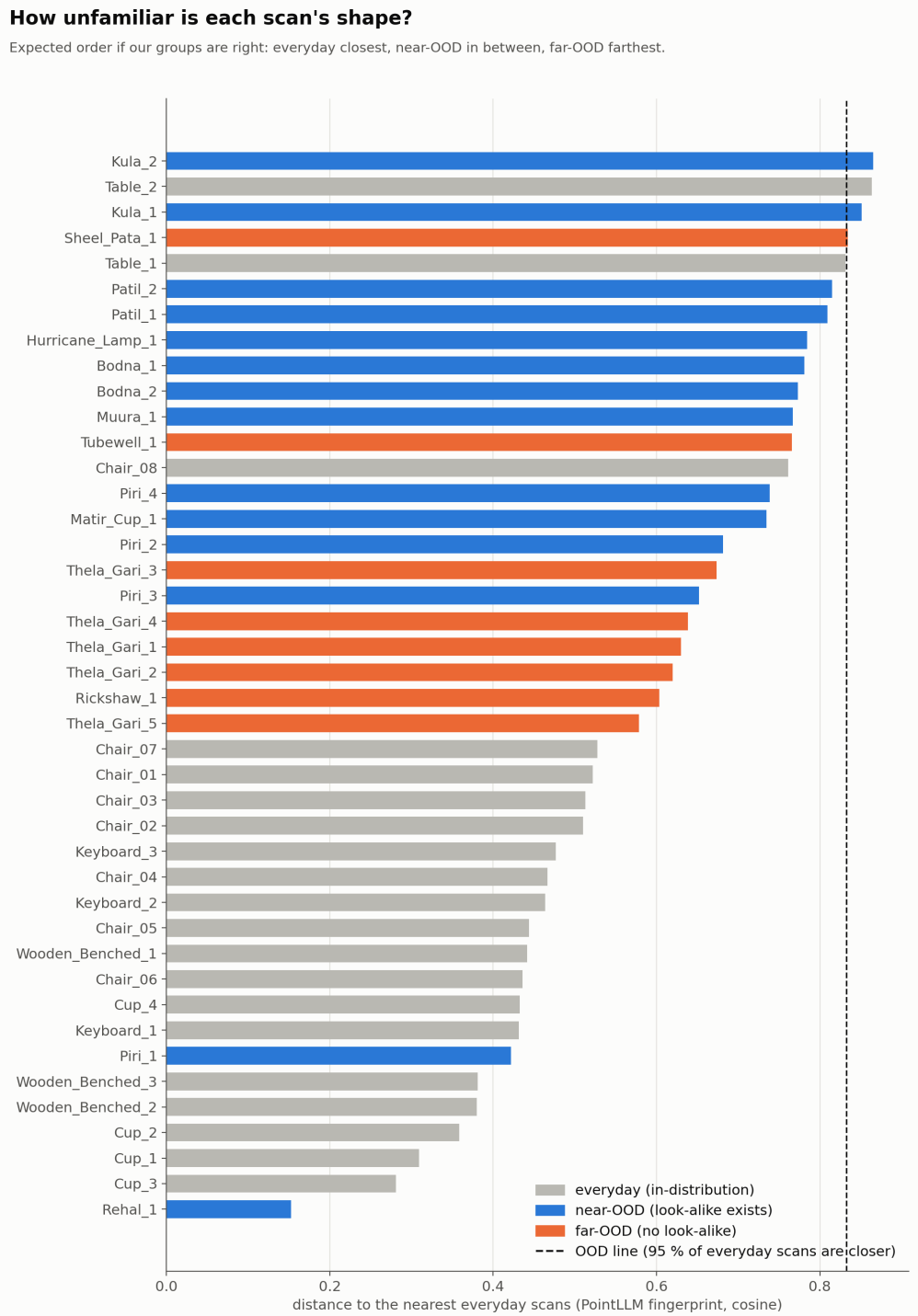


tubewell (far_ood): what the model sees

3. Are the groups really different? (OOD check)



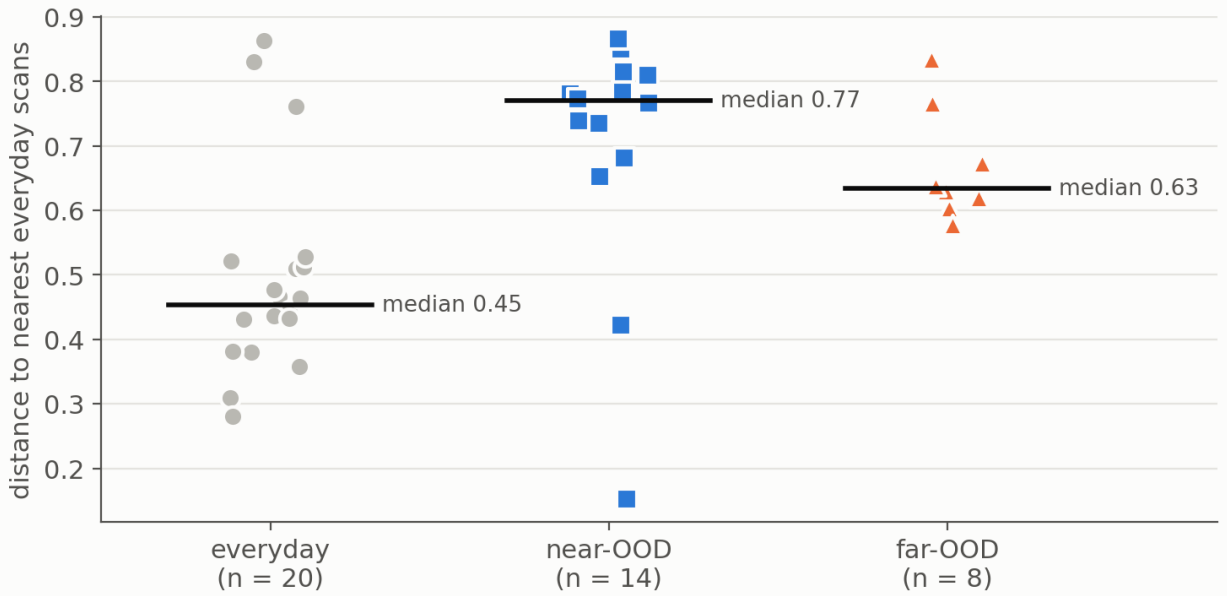
Training captions containing each word (of 658,577).



Distance to the 3 nearest everyday scans. OOD line = 0.833.

Distance by group

Each dot is one scan; the black line is the group median. Expected: everyday < near-OOD < far-OOD.

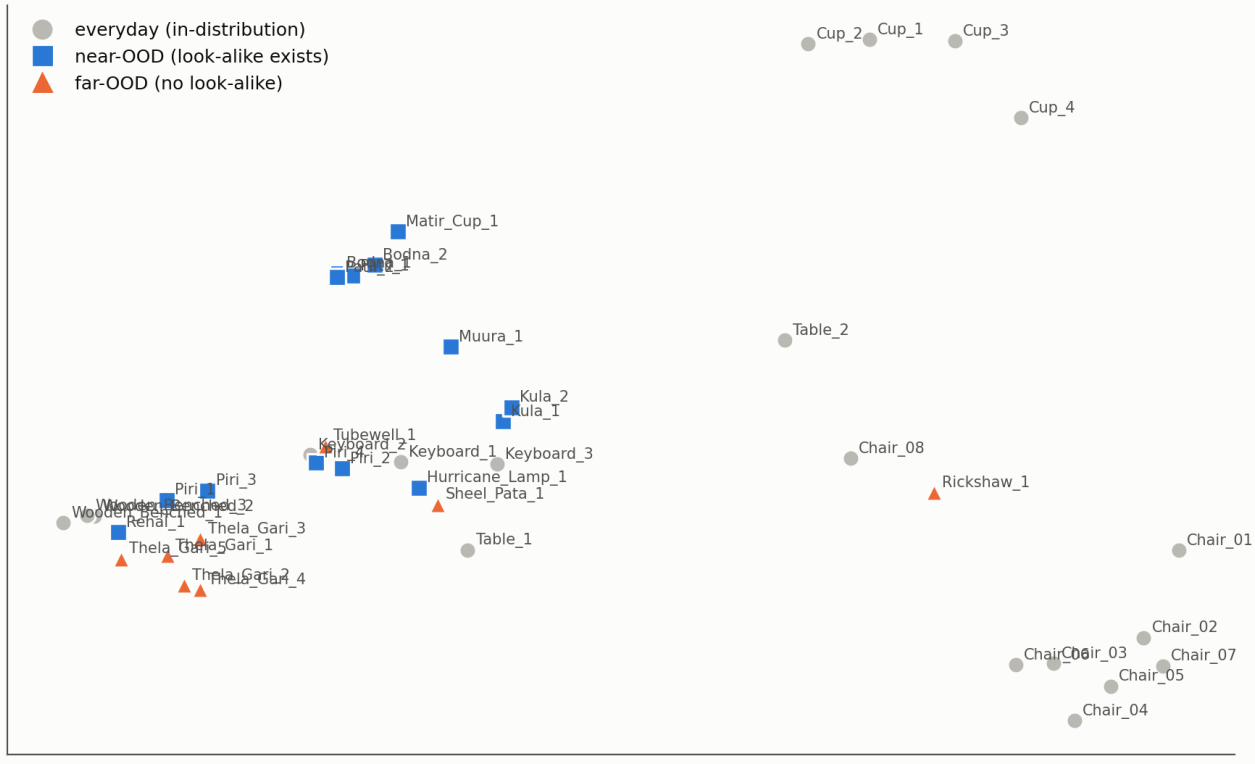


The same distances by group; black line = median.

group	count	median	min	max
everyday	20	0.454	0.281	0.864
near_ood	14	0.77	0.153	0.865
far_ood	8	0.634	0.578	0.834

Map of PointLLM's shape fingerprints

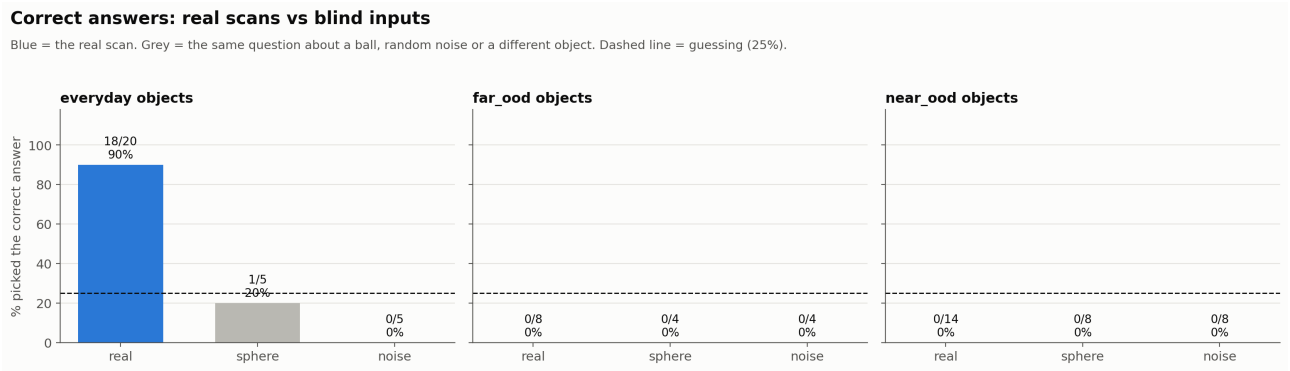
Close points = PointLLM sees the shapes as similar (PCA of the 384-number fingerprints).



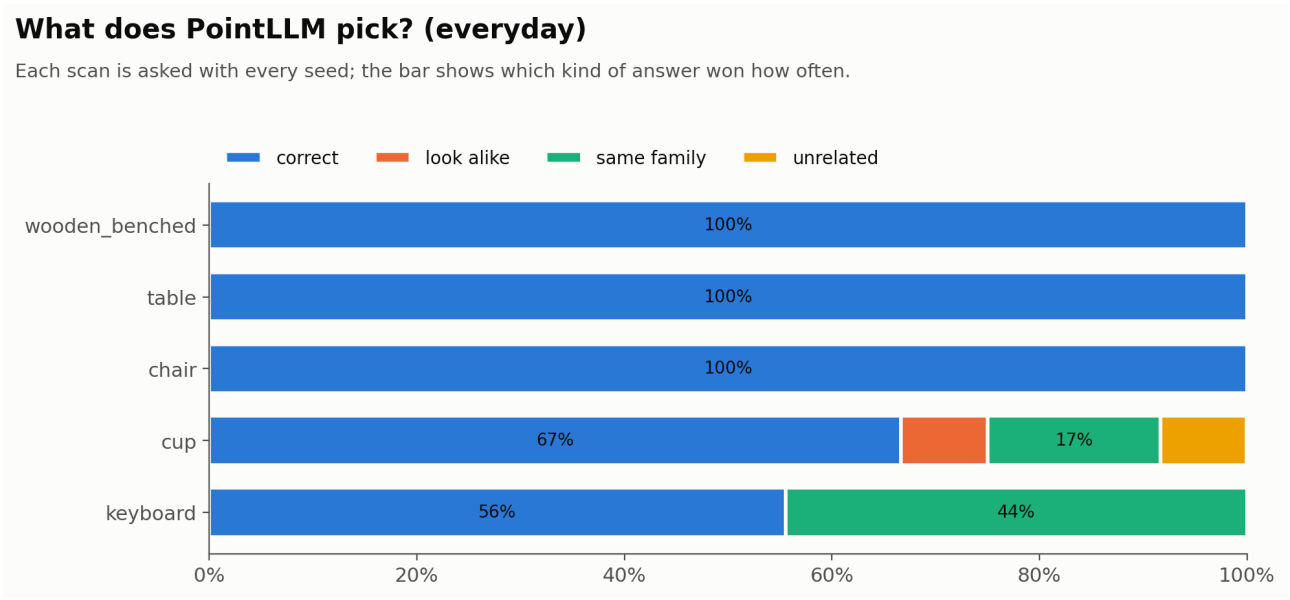
Map of PointLLM's fingerprints (PCA).

4. Multiple choice (each object's own 4 options)

set	label	correct	n	accuracy	p_value
everyday	real scans	18	20	0.9	0.0
everyday	real, per object (majority)	5	5	1.0	0.001
everyday	blind: sphere	1	5	0.2	0.763
everyday	blind: noise	0	5	0.0	1.0
far_ood	real scans	0	8	0.0	1.0
far_ood	real, per object (majority)	0	4	0.0	1.0
far_ood	blind: sphere	0	4	0.0	1.0
far_ood	blind: noise	0	4	0.0	1.0
near_ood	real scans	0	14	0.0	1.0
near_ood	real, per object (majority)	0	8	0.0	1.0
near_ood	blind: sphere	0	8	0.0	1.0
near_ood	blind: noise	0	8	0.0	1.0



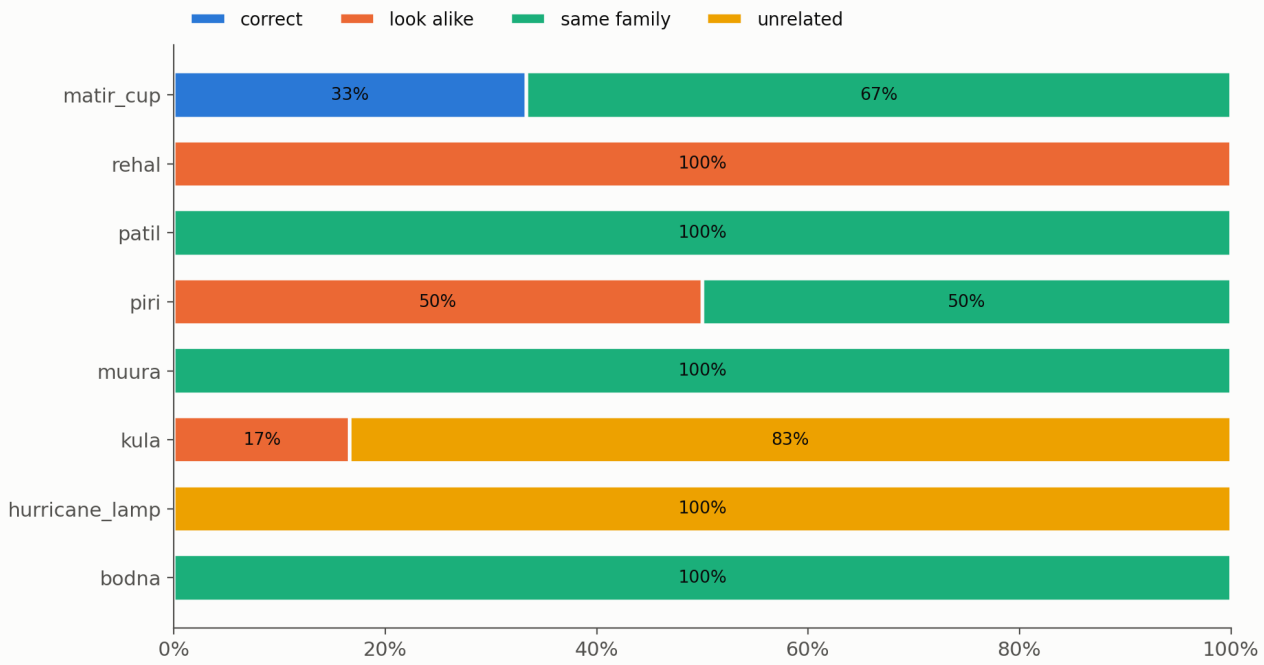
Real scans vs ball and noise. Dashed line = guessing (25 %).



Which kind of answer wins (everyday).

What does PointLLM pick? (near_ood)

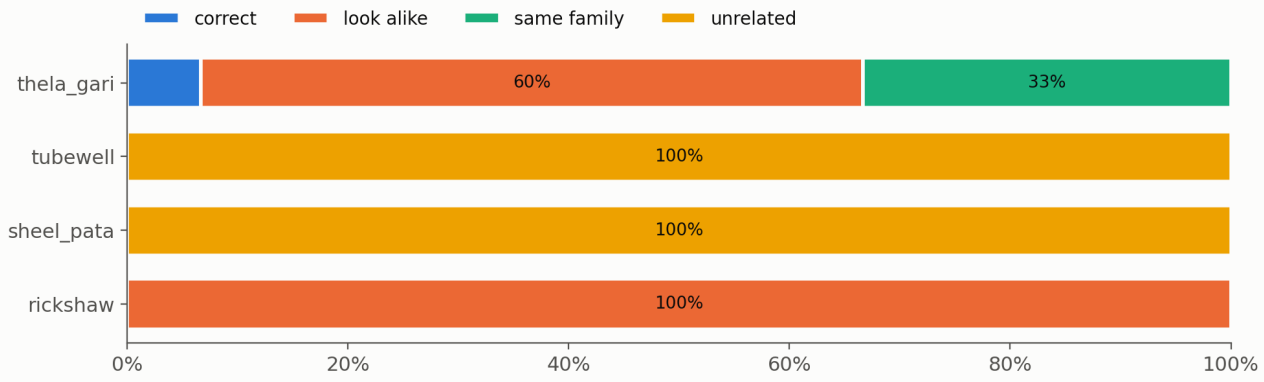
Each scan is asked with every seed; the bar shows which kind of answer won how often.



Which kind of answer wins (near_ood).

What does PointLLM pick? (far_ood)

Each scan is asked with every seed; the bar shows which kind of answer won how often.



Which kind of answer wins (far_ood).

set	file	pick_text	pick	seeds_agreeing
everyday	Chair_01_cleaned.ply	chair	correct	3
everyday	Chair_02_cleaned.ply	chair	correct	3
everyday	Chair_03_cleaned.ply	chair	correct	3
everyday	Chair_04_cleaned.ply	chair	correct	3
everyday	Chair_05_cleaned.ply	chair	correct	3
everyday	Chair_06_cleaned.ply	chair	correct	3
everyday	Chair_07_cleaned.ply	chair	correct	3
everyday	Chair_08_cleaned.ply	chair	correct	3
everyday	Cup_1_cleaned.ply	cup	correct	3
everyday	Cup_2_cleaned.ply	cup	correct	3
everyday	Cup_3_cleaned.ply	cup	correct	2

set	file	pick_text	pick	seeds_agreeing
everyday	Cup_4_cleaned.ply	chair	unrelated	1
everyday	Keyboard_1_cleaned.ply	laptop	same_family	3
everyday	Keyboard_2_cleaned.ply	keyboard	correct	3
everyday	Keyboard_3_cleaned.ply	keyboard	correct	2
everyday	Table_1_cleaned.ply	table	correct	3
everyday	Table_2_cleaned.ply	table	correct	3
everyday	Wooden_Benched_1_cleaned.ply	bench	correct	3
everyday	Wooden_Benched_2_cleaned.ply	bench	correct	3
everyday	Wooden_Benched_3_cleaned.ply	bench	correct	3
far_ood	Rickshaw_1.ply	tricycle	look_alike	3
far_ood	Sheel_Pata_1_cleaned.ply	chair	unrelated	3
far_ood	Thela_Gari_1_cleaned.ply	wagon	same_family	3
far_ood	Thela_Gari_2_cleaned.ply	table	look_alike	3
far_ood	Thela_Gari_3_cleaned.ply	wagon	same_family	2
far_ood	Thela_Gari_4_cleaned.ply	table	look_alike	3
far_ood	Thela_Gari_5_cleaned.ply	table	look_alike	3
far_ood	Tubewell_1_cleaned.ply	chair	unrelated	3
near_ood	Bodna_1_cleaned.ply	vase	same_family	3
near_ood	Bodna_2_cleaned.ply	vase	same_family	3
near_ood	Hurricane_Lamp_1_cleaned.ply	chair	unrelated	3
near_ood	Kula_1_cleaned.ply	chair	unrelated	2
near_ood	Kula_2_cleaned.ply	chair	unrelated	3
near_ood	Matir_Cup_1_cleaned.ply	bowl	same_family	2
near_ood	Muura_1_cleaned.ply	chair	same_family	3
near_ood	Patil_1_cleaned.ply	bowl	same_family	3
near_ood	Patil_2_cleaned.ply	bowl	same_family	3
near_ood	Piri_1_cleaned.ply	bench	same_family	3
near_ood	Piri_2_cleaned.ply	bench	same_family	3
near_ood	Piri_3_cleaned.ply	box	look_alike	3
near_ood	Piri_4_cleaned.ply	box	look_alike	3
near_ood	Rehal_1_cleaned.ply	bench	look_alike	3

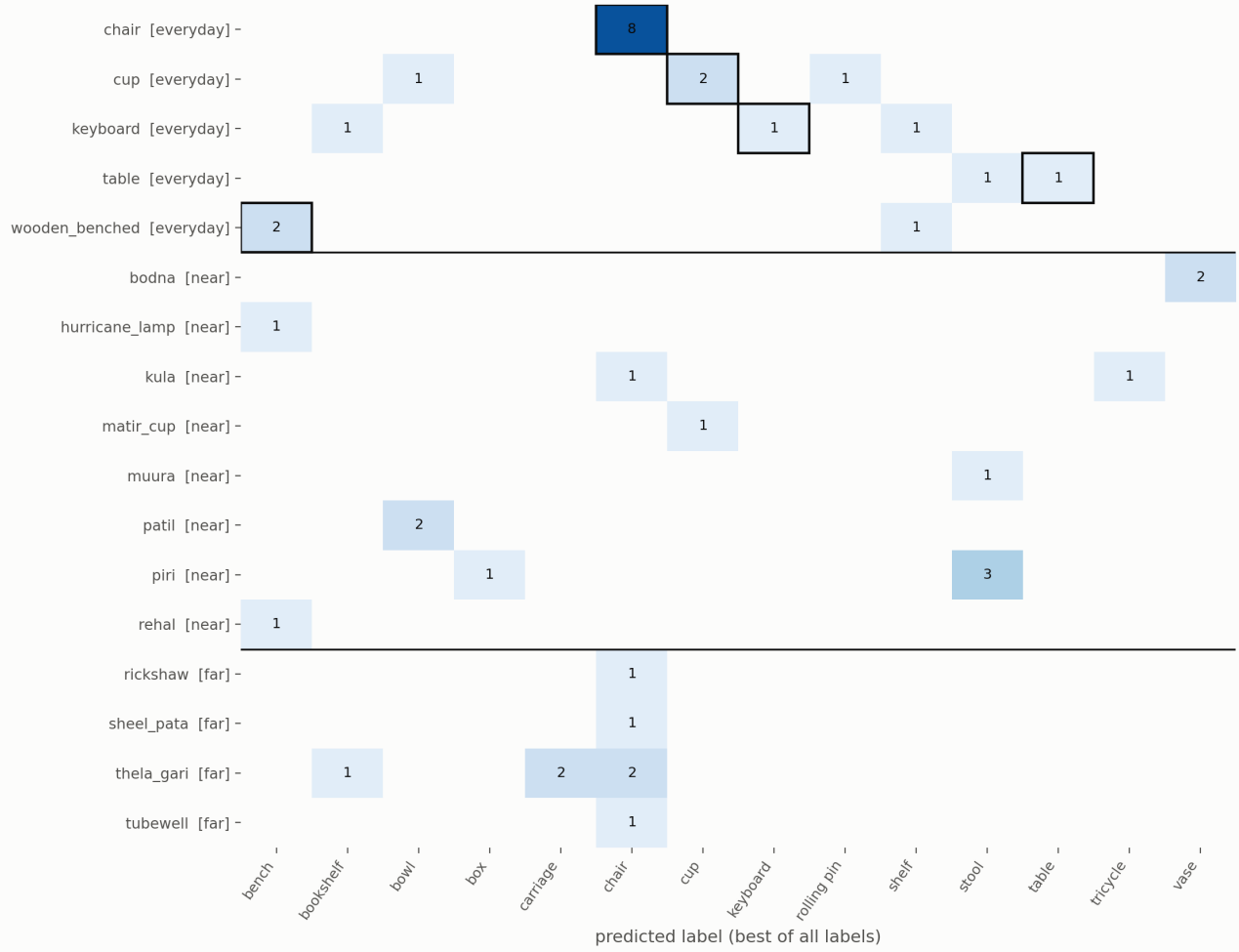
5. All-labels test and confusion matrix

group	scans	top1	top3	score
everyday	20	14	17	plain
near_ood	14	0	2	plain
far_ood	8	0	0	plain

group	scans	top1	top3	score
everyday	20	5	12	calibrated
near_ood	14	1	5	calibrated
far_ood	8	0	0	calibrated

Confusion matrix (plain): what PointLLM calls each object

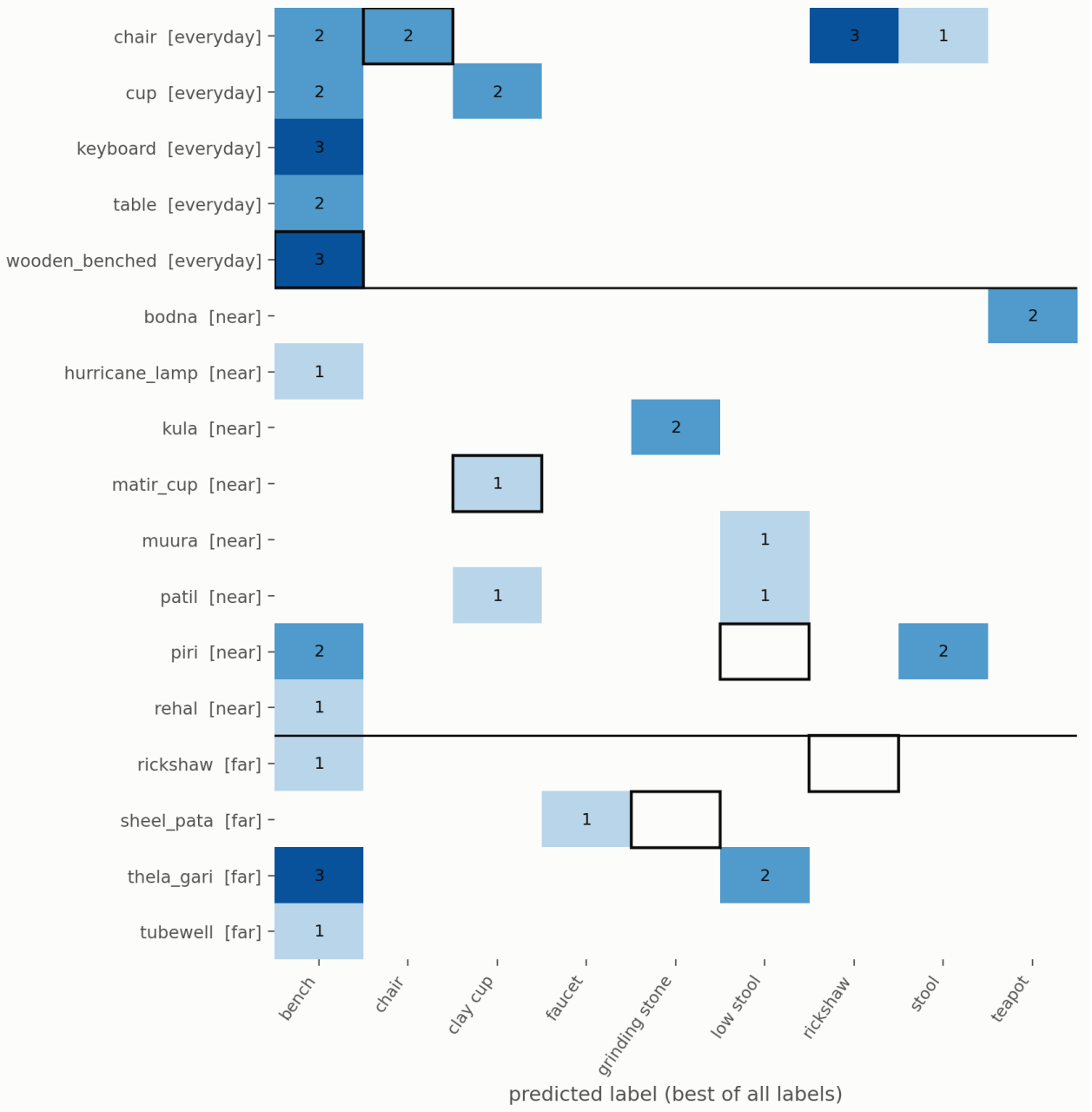
Cells = number of scans. Black outline = the correct label for that row. Rows are split into everyday / near / far.



Plain likelihood. Outline = correct label.

Confusion matrix (calibrated): what PointLLM calls each object

Cells = number of scans. Black outline = the correct label for that row. Rows are split into everyday / near / far.



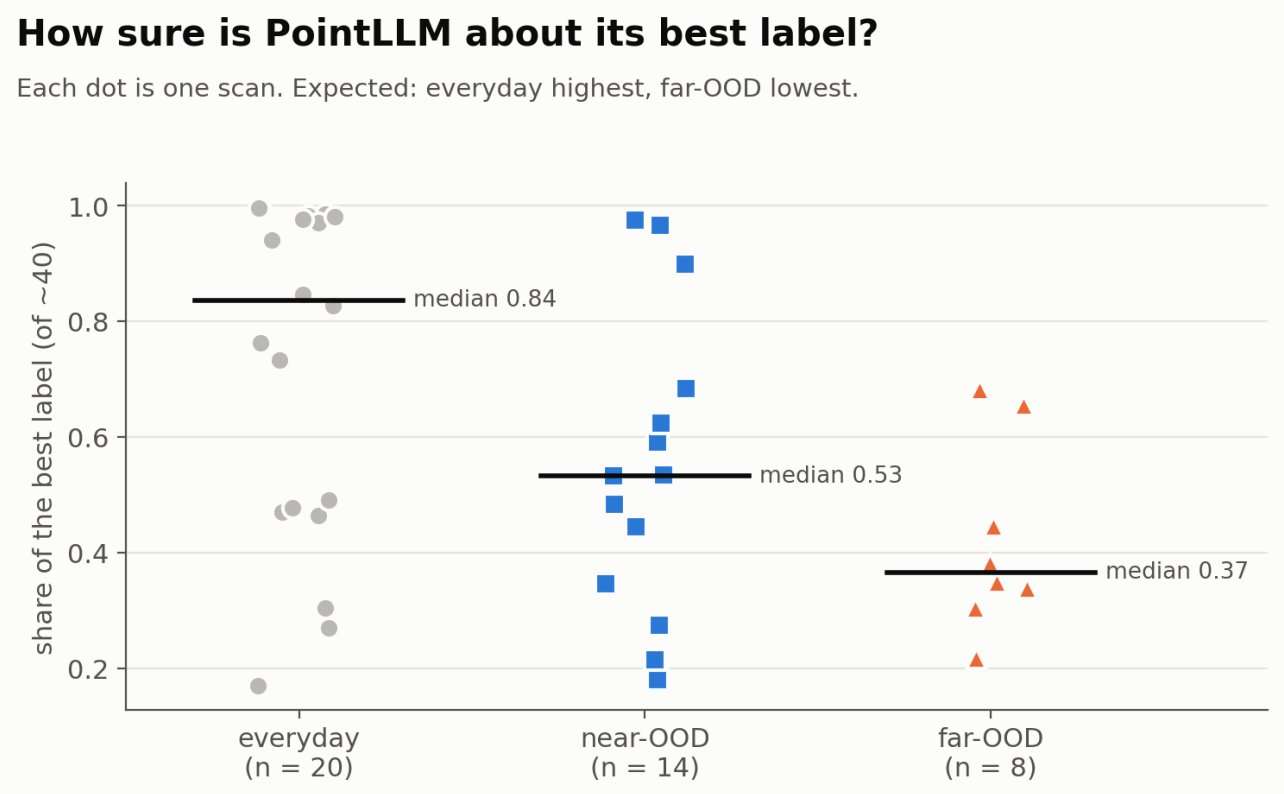
Calibrated (minus ball/noise). Outline = correct label.

group	file	predicted	top3	correct_label
everyday	Chair_01_cleaned.ply	bench	bench, chair, rickshaw	chair
everyday	Chair_02_cleaned.ply	stool	stool, chair, rickshaw	chair
everyday	Chair_03_cleaned.ply	rickshaw	rickshaw, chair, bench	chair
everyday	Chair_04_cleaned.ply	rickshaw	rickshaw, low stool, chair	chair
everyday	Chair_05_cleaned.ply	chair	chair, stool, bench	chair
everyday	Chair_06_cleaned.ply	bench	bench, rickshaw, book stand	chair
everyday	Chair_07_cleaned.ply	rickshaw	rickshaw, bench, chair	chair
everyday	Chair_08_cleaned.ply	chair	chair, rickshaw, bench	chair

group	file	predicted	top3	correct_label
everyday	Cup_1_cleaned.ply	clay cup	clay cup, faucet, bench	cup
everyday	Cup_2_cleaned.ply	clay cup	clay cup, faucet, bench	cup
everyday	Cup_3_cleaned.ply	bench	bench, grinding stone, clay cup	cup
everyday	Cup_4_cleaned.ply	bench	bench, tray, faucet	cup
everyday	Keyboard_1_cleaned.ply	bench	bench, grinding stone, cutting board	keyboard
everyday	Keyboard_2_cleaned.ply	bench	bench, keyboard, shelf	keyboard
everyday	Keyboard_3_cleaned.ply	bench	bench, grinding stone, cutting board	keyboard
everyday	Table_1_cleaned.ply	bench	bench, table, book stand	table
everyday	Table_2_cleaned.ply	bench	bench, stool, low stool	table
everyday	Wooden_Benched_1_cleaned.ply	bench	bench, shelf, low stool	bench
everyday	Wooden_Benched_2_cleaned.ply	bench	bench, low stool, stool	bench
everyday	Wooden_Benched_3_cleaned.ply	bench	bench, low stool, shelf	bench
far_ood	Rickshaw_1.ply	bench	bench, grinding stone, faucet	rickshaw
far_ood	Sheel_Pata_1_cleaned.ply	faucet	faucet, bench, low stool	grinding stone
far_ood	Thela_Gari_1_cleaned.ply	bench	bench, rickshaw, low stool	push cart
far_ood	Thela_Gari_2_cleaned.ply	low stool	low stool, bench, rickshaw	push cart
far_ood	Thela_Gari_3_cleaned.ply	bench	bench, rickshaw, low stool	push cart
far_ood	Thela_Gari_4_cleaned.ply	low stool	low stool, bench, rickshaw	push cart
far_ood	Thela_Gari_5_cleaned.ply	bench	bench, low stool, grinding stone	push cart
far_ood	Tubewell_1_cleaned.ply	bench	bench, faucet, rickshaw	hand pump
near_ood	Bodna_1_cleaned.ply	teapot	teapot, faucet, bench	water jug
near_ood	Bodna_2_cleaned.ply	teapot	teapot, low stool, faucet	water jug
near_ood	Hurricane_Lamp_1_cleaned.ply	bench	bench, faucet, book stand	lantern
near_ood	Kula_1_cleaned.ply	grinding stone	grinding stone, cutting board, bench	bamboo tray
near_ood	Kula_2_cleaned.ply	grinding stone	grinding stone, bench, cutting board	bamboo tray
near_ood	Matir_Cup_1_cleaned.ply	clay cup	clay cup, bench, cup	clay cup
near_ood	Muura_1_cleaned.ply	low stool	low stool, stool, bench	cane stool
near_ood	Patil_1_cleaned.ply	clay cup	clay cup, teapot, bamboo tray	cooking pot
near_ood	Patil_2_cleaned.ply	low stool	low stool, bench, clay cup	cooking pot
near_ood	Piri_1_cleaned.ply	bench	bench, low stool, stool	low stool
near_ood	Piri_2_cleaned.ply	stool	stool, low stool, bench	low stool
near_ood	Piri_3_cleaned.ply	bench	bench, low stool, tray	low stool
near_ood	Piri_4_cleaned.ply	stool	stool, bench, low stool	low stool

group	file	predicted	top3	correct_label
near_ood	Rehal_1_cleaned.ply	bench	bench, low stool, stool	book stand

6. How sure is PointLLM? (confidence)



Share of the best label among all labels; lower = less sure.

group	count	median	min	max
everyday	20	0.837	0.169	0.997
near_ood	14	0.533	0.18	0.976
far_ood	8	0.366	0.218	0.682

7. How to read this

- **Far-OOD is confirmed** if far-OOD scans are farthest from the everyday scans, PointLLM is least sure about them, and its labels for them are scattered or unrelated.
- **Near-OOD** objects should go to their look-alike (rehal -> bench), with the shape still pointing that way after calibration.
- Small groups (3 far-OOD scans): read the dots, not just the medians. One scan can move a median.
- Raw tables are in tables/; every chart is in figures/.